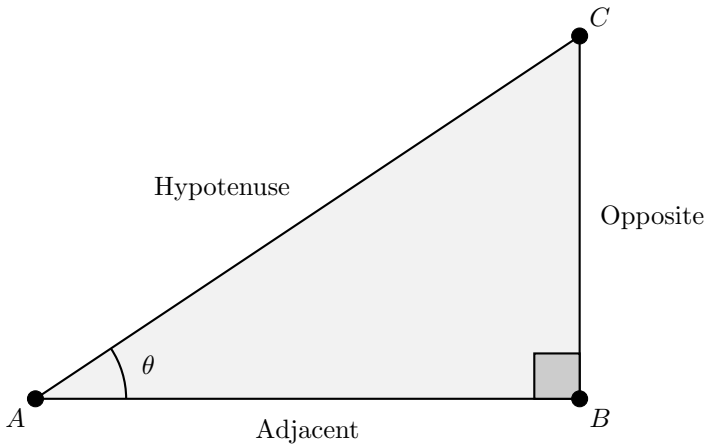
1 Foundational Definitions



Let $\triangle ABC$ be a right triangle with $\angle B = 90°$, and let $\theta = \angle BAC$ denote the angle at vertex A. Thus, the hypotenuse is the side AC, and the segment AB is adjacent and BC opposite to θ. The trigonometric functions are defined as the following:

Definition 1 (Sine)**.**

$$\sin\theta = \frac{BC}{AC} = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

Definition 2 (Cosecant)**.**

$$\csc\theta = \frac{AC}{BC} = \frac{\text{Hypotenuse}}{\text{Opposite}}$$

Definition 3 (Cosine)**.**

$$\cos\theta = \frac{AB}{AC} = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

Definition 4 (Secant)**.**

$$\sec\theta = \frac{AC}{AB} = \frac{\text{Hypotenuse}}{\text{Adjacent}}$$

Definition 5 (Tangent)**.**

$$\tan\theta = \frac{BC}{AB} = \frac{\text{Opposite}}{\text{Adjacent}}$$

Definition 6 (Cotangent)**.**

$$\cot\theta = \frac{AB}{BC} = \frac{\text{Adjacent}}{\text{Opposite}}$$

2 Immediate Algebraic Properties

2.1 Reciprocal Identities

2.2 Quotient Identities

3 Cofunction Identities

Definition 7 (Complementary Angles)**.** Two angles are *complementary* if and only if the sum of their measures is equal to the measure of a right angle (90° or $\frac{\pi}{2}$ radians).

Formally, let α and β be the measures of two angles. They are complementary if and only if: $\alpha + \beta = 90°$. Either angle is referred to as the complement of the other.

Theorem 1 (Congruence of Complements)**.** *If two angles are complementary to the same angle, then they are congruent to each other. Similarly, if they are complementary to two congruent angles, they are congruent.*

Proof. wasd □

Theorem 2 (Complementarity of Acute Angles in a Right Triangle)**.** *The two acute angles of any right-angled triangle are complementary.*

Proof. wasd □

Identity 1 (Cofunction Identities)**.**

$$\sin(90^\circ - \theta) = \cos\theta, \quad \cos(90^\circ - \theta) = \sin\theta$$

Proof. wasd □

4 Pythagorean Identities

Theorem 3 (Pythagorean Theorem)**.** *For right triangle $\triangle ABC$, the lengths of the sides satisfy:*

$$AB^2 + BC^2 = AC^2$$

Identity 2 (Pythagorean Identity)**.**

$$\sin^2\theta + \cos^2\theta = 1$$

Proof. Using the side-length definitions of sine and cosine for $\triangle ABC$:

$$\sin\theta = \frac{BC}{AC} \quad \text{and} \quad \cos\theta = \frac{AB}{AC}$$

Squaring both terms gives:

$$\sin^2\theta = \frac{BC^2}{AC^2} \quad \text{and} \quad \cos^2\theta = \frac{AB^2}{AC^2}$$

Adding them together:

$$\sin^2\theta + \cos^2\theta = \frac{BC^2}{AC^2} + \frac{AB^2}{AC^2} = \frac{AB^2 + BC^2}{AC^2}$$

By the Pythagorean Theorem, $AB^2 + BC^2 = AC^2$, which yields:

$$\sin^2\theta + \cos^2\theta = \frac{AC^2}{AC^2} = 1$$

□

Identity 3 (Pythagorean Identity)**.**

$$\sec^2\theta - \tan^2\theta = 1$$

Proof. From the Pythagorean Identity, divide both sides by $\cos^2\theta$, yielding

$$\begin{aligned} \sin^2\theta + \cos^2\theta = 1 &\implies \frac{\sin^2\theta}{\cos^2\theta} + \frac{\cos^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta} \\ &\implies \frac{\sin^2\theta}{\cos^2\theta} + 1 = \frac{1}{\cos^2\theta} \end{aligned}$$

By the definitions of tangent and secant, we have

$$\begin{aligned} \frac{\sin^2\theta}{\cos^2\theta} + 1 = \frac{1}{\cos^2\theta} &\implies \tan^2\theta + 1 = \sec^2\theta \\ &\implies \sec^2\theta - \tan^2\theta = 1 \end{aligned}$$

□

Identity 4 (Pythagorean Identity)**.**

$$\csc^2\theta - \cot^2\theta = 1$$

Proof. From the Pythagorean Identity, divide both sides by $\sin^2\theta$, yielding

$$\begin{aligned}\sin^2\theta + \cos^2\theta = 1 &\implies \frac{\sin^2\theta}{\sin^2\theta} + \frac{\cos^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta} \\ &\implies 1 + \frac{\cos^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta}\end{aligned}$$

By the definition of cotangent and cosecant, we have

$$\begin{aligned}1 + \frac{\cos^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta} &\implies 1 + \cot^2\theta = \csc^2\theta \\ &\implies \csc^2\theta - \cot^2\theta = 1\end{aligned}$$

□

5 Angle Sum and Difference

6 Double-Angle Identities

7 Half-Angle and Power Reducing

8 Even/Odd Identities